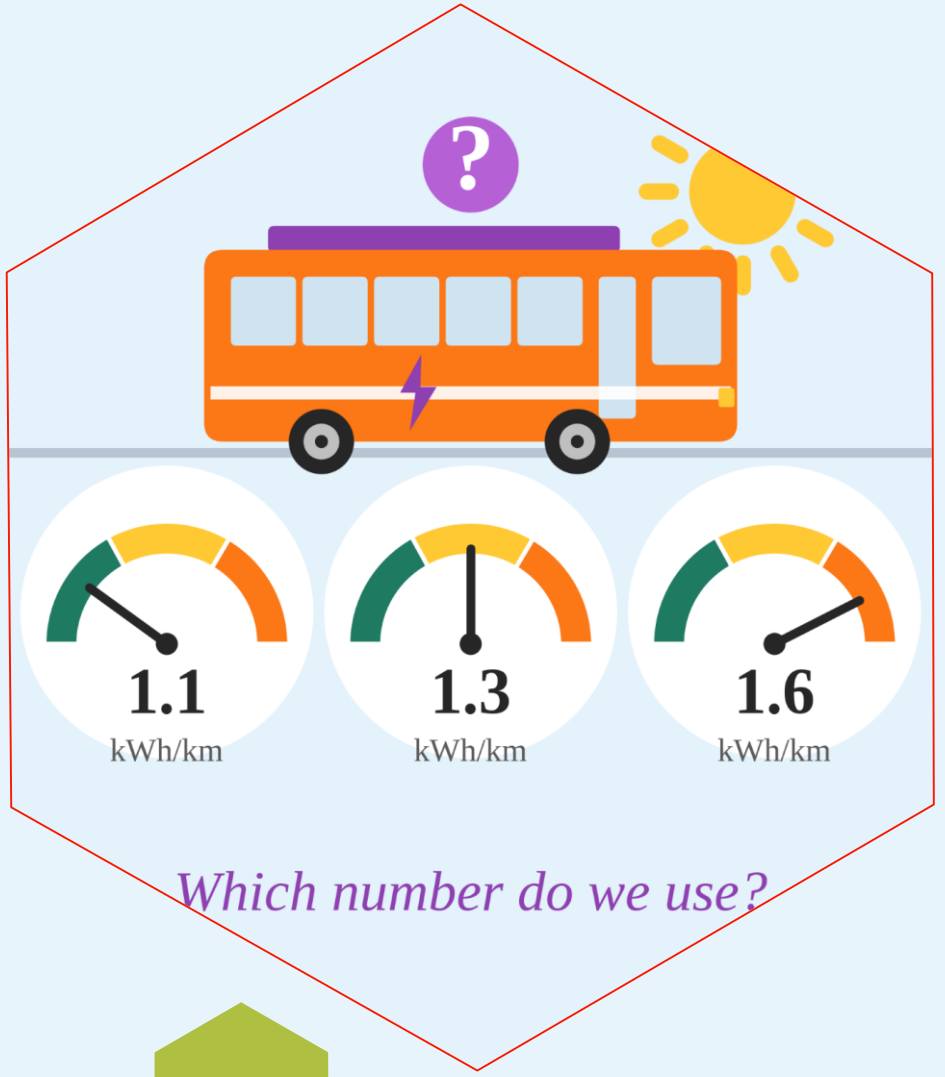
Lecture 7:
Quantitative Reasoning II

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Disclaimer: this lecture references material adapted from Kosky, Balmer, Keat & Wise, Exploring Engineering (5th ed.), and from the instructor's own course slides, with modifications. No authorship of third-party material is claimed.

Last Week, in One Breath

Demand and capacity — what is asked against what can be delivered, with units and a time basis.

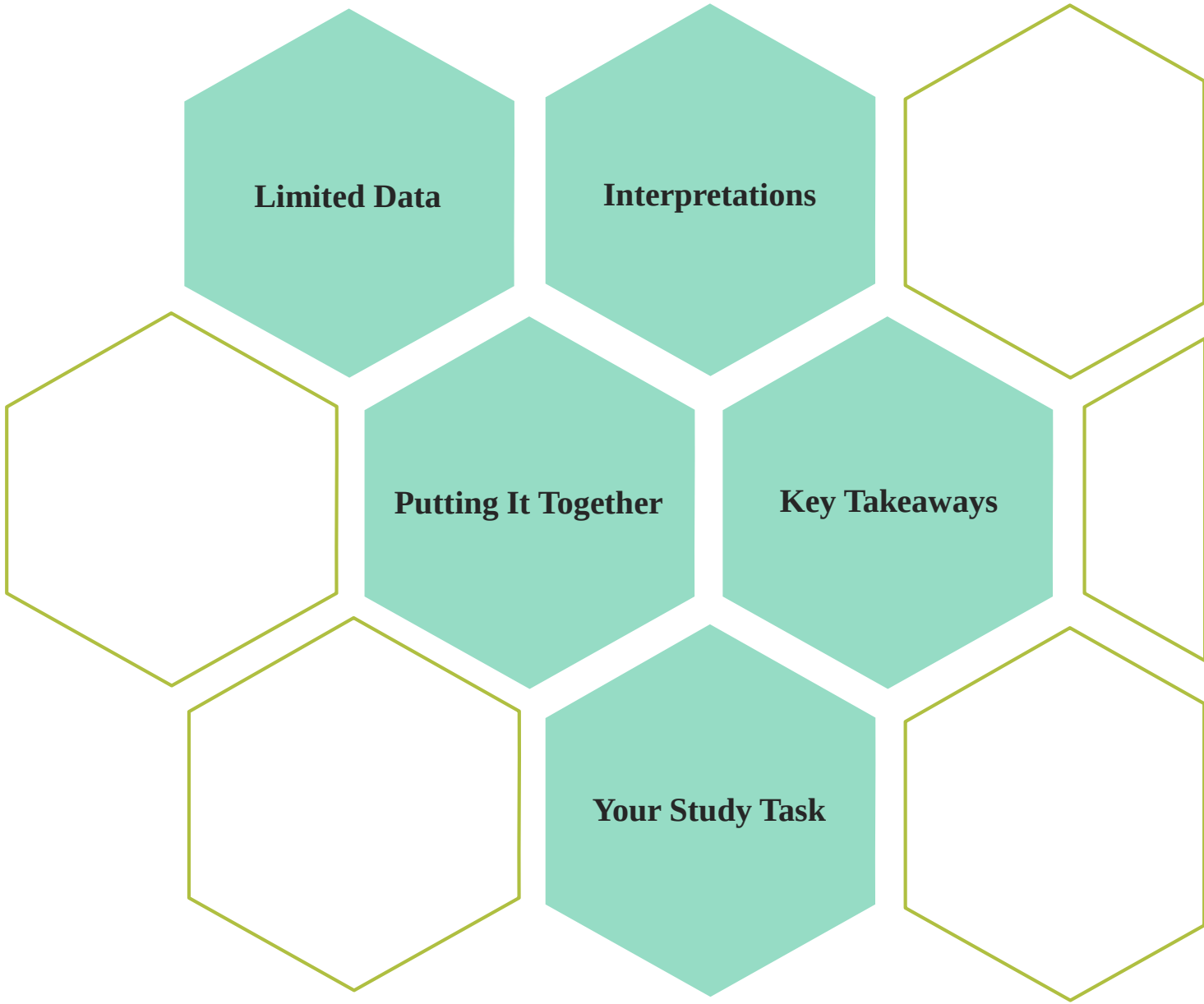
Rates and flows — quantity per time. Power × time = energy. Change = in − out.

Estimation — break into factors, round to one digit, check the units.

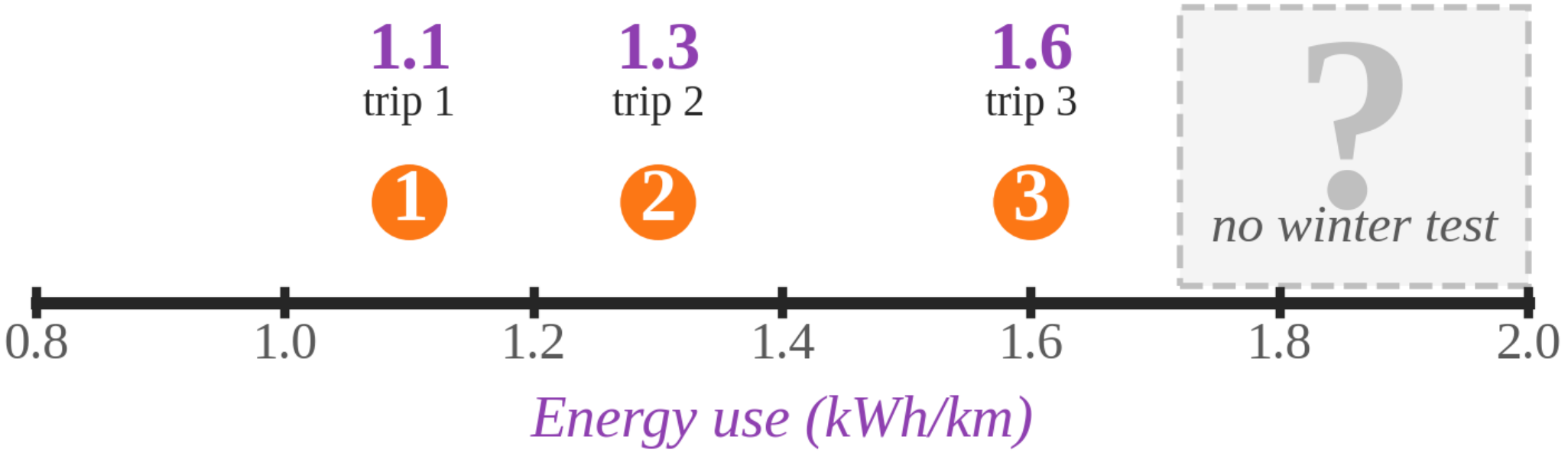
Bounding — low, best, and high; stop when the decision does not change.

Today — reason when data is limited, and read the same numbers more than one way.

Agenda

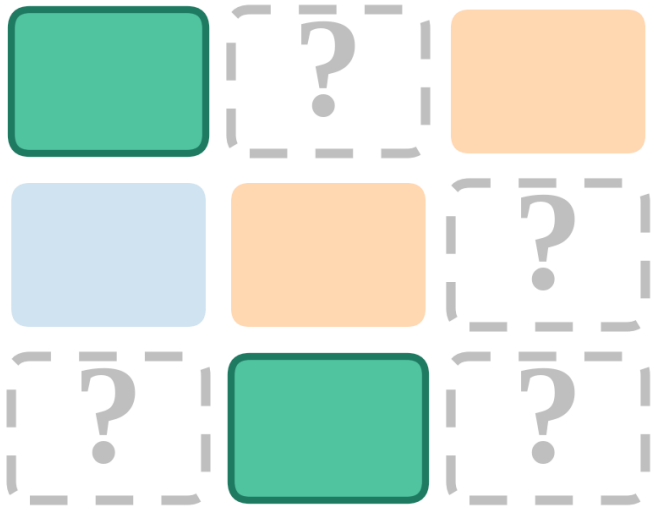

Limited Data
Interpretations
Putting It Together
Key Takeaways
Your Study Task

Three Trips, Three Numbers



Three test trips: 1.1, 1.3, and 1.6 kWh/km. ***Which number do we use to promise a range?***

PART 1



some of the data is missing

Limited and Incomplete Data

What to do when the numbers you need are not there.

Five Ways Data Can Be Incomplete

Problem	What it means	Bus example
Too little	A few data points only	Three test trips
Missing	No data for the case we care about	No winter test
Noisy	Values scatter around the truth	1.1 to 1.6 kWh/km on the same route
Out of date or unrepresentative	Measured under other conditions	Tests with an empty bus
Wrong definition or unit	Quantity not what we think	Battery “%” or “200” with no unit

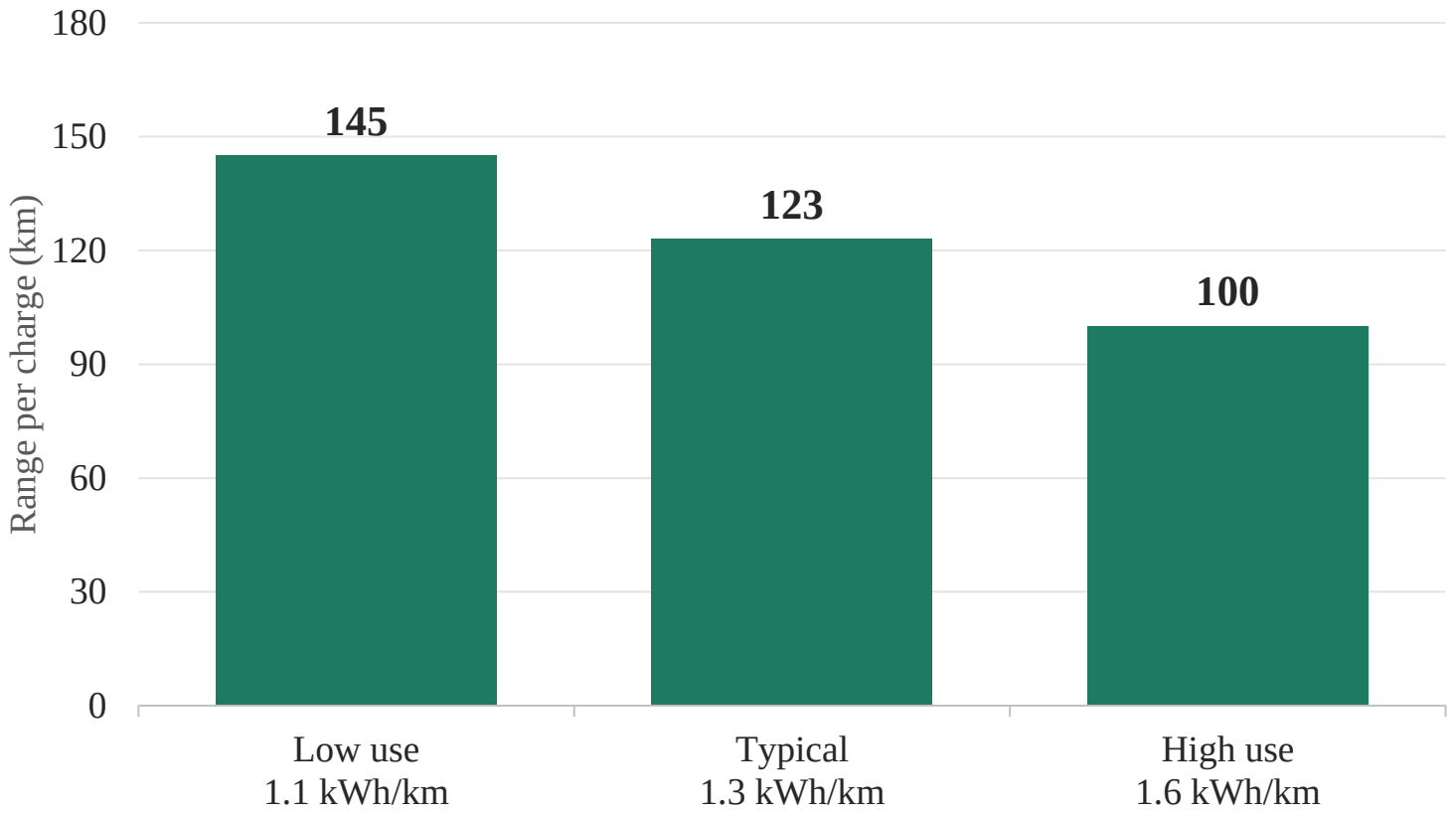
Name the problem first. The fix depends on which one it is.

Where Numbers Come From

Source	Example	How far to trust it
Measure	Time a bus trip, log the charger	High, if the method is stated
Look up	A datasheet, a standard	Medium: conditions may differ
Ask	The driver, the facilities manager	Medium: check it against something
Estimate	Break into factors (Lecture 6)	Low to medium: state assumptions
Analogy	A similar building or route	Low: say how similar it is

Write the source and your confidence next to every number.

Use Ranges, Not Single Numbers



DAILY NEED

96 km

All three cases clear it.

The high-use case clears it by only 4 km.

Report: "about 100 to 145 km".

A range tells the reader how far to trust the answer.

Illustrative values. Range = 160 kWh ÷ energy use per km.

Small Samples: Say How Many

MEAN

1.33 kWh/km

(1.1 + 1.3 + 1.6) ÷ 3. One odd trip moves it a lot.

MEDIAN

1.3 kWh/km

The middle value. One odd trip barely moves it.

SAMPLE SIZE

n = 3

Write n next to every number you report.

With three points, give the range and n. Do not claim more than three trips can show.

Significant figures and rough data: Kosky et al., Ch. 2 (§2.5). Illustrative values.

Triangulate: Estimate It Three Ways

≈ 2,800

kWh per day

8,000 m² × 0.35 [kWh/m²-day]

≈ 2,260

kWh per day

94 [kW] × 24 [h]

≈ 2,330

kWh per day

70,000 [kWh/month] ÷ 30

Route 1: area × typical use. **Route 2:** Lecture 6 load profile. **Route 3:** monthly bill.

Three independent routes agree within about 25%. Agreement builds trust; disagreement tells you where to look.

Illustrative values. Checking an estimate against a second source: Kosky et al., Ch. 2 (Example 2.4).

Name the Problem

On your own, on paper: name the problem type for each statement, and write one way to get a better number.

A “We tested the bus on one trip.”

B “There is no data on the dormitory’s winter heating.”

C “Five readings of one charger: 58, 61, 54, 63, 60 kW.”

D “The solar yield figure comes from a desert site.”

E “The datasheet says ‘200’ with no unit.”

F “The load profile is from before the new wing opened.”

Five minutes, on your own. Use the five problem types from earlier.

Name the Problem: Answer Key

	Problem	A better number
A	Too little	Repeat the trip on several days and loads.
B	Missing	Estimate by analogy with other buildings; plan a winter measurement.
C	Noisy	Report the mean and range: about 59 kW, from 54 to 63.
D	Unrepresentative	Find yield data for a similar climate.
E	Wrong definition or unit	Check the label: 200 kWh is energy, 200 kW is power.
F	Out of date	Re-measure, or scale the profile for the new wing.

The problem type tells you which fix to try.

PART 2

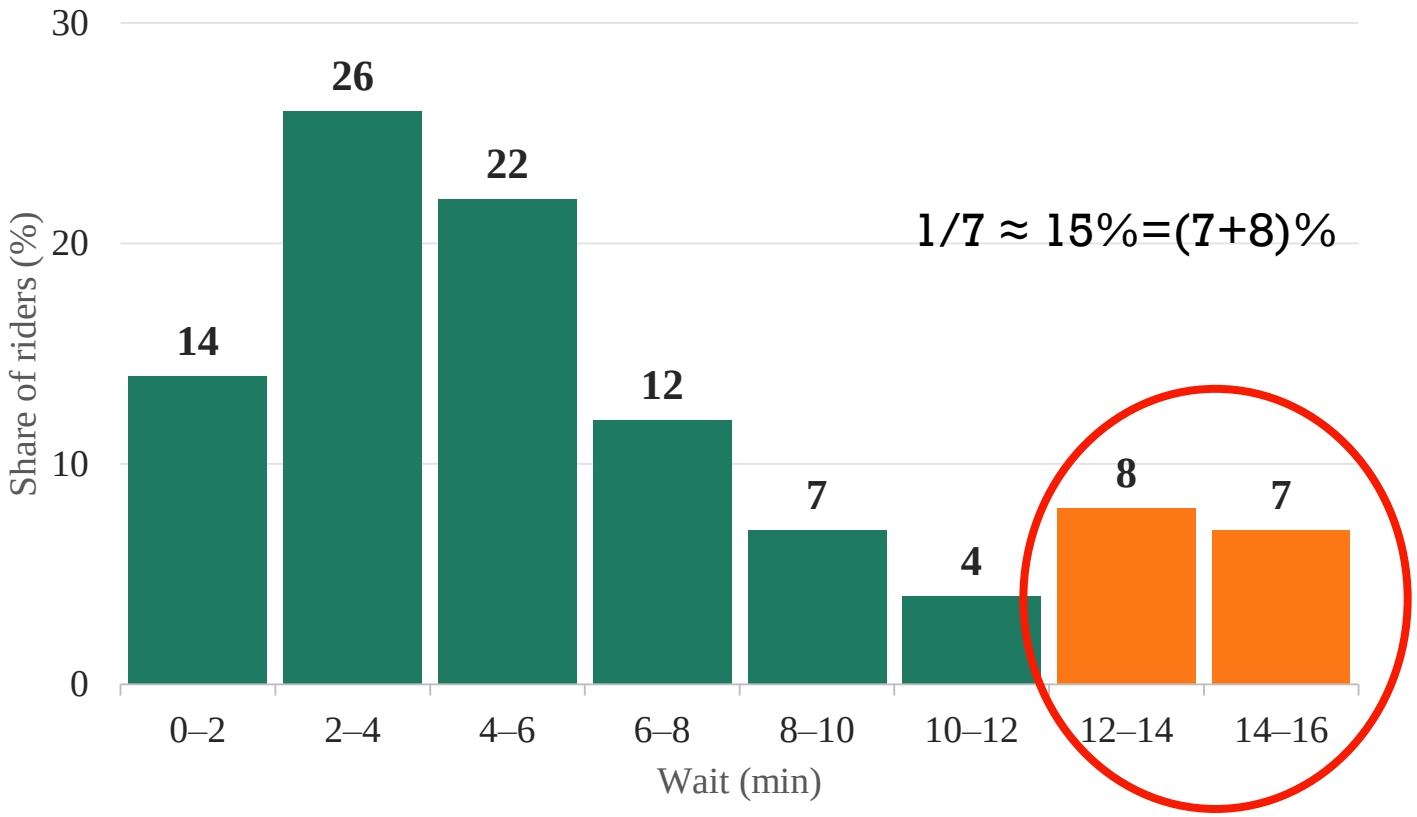


one data set, two readings

Comparing Interpretations

The same numbers can support different readings.

Same Data, Different Readings



DISPATCHER

"Average wait is 6 minutes. Good."

RIDER

"One in seven of us waits over 12 minutes."

ADMINISTRATOR

"Cutting the long waits needs more buses. What does that cost?"

Same numbers, three readings. All three are true.

Illustrative wait-time distribution, not measured data.

Choose the Metric for the Decision

Question	Metric	Value here
How long does a typical rider wait?	Median	about 5 min
How much waiting is there in total?	Average	about 6 min × riders
How bad is it for the unlucky?	90th percentile	about 13 min
What is the worst case?	Maximum	about 16 min

The decision picks the metric, not the other way round.

Lecture 6 had the same choice: kWh answers "how much?", kW answers "how fast?".

Illustrative values from the previous slide's distribution.

The Baseline Changes the Story

60%

of the corridor lighting

12 kW ÷ 20 kW

7%

of the building’s peak

12 kW ÷ 180 kW

One fact: an LED retrofit saves 12 kW. Two true percentages.

Always ask: a percentage of what? Different baselines and boundaries give different, true stories.

Illustrative values.

Misleading Comparisons

Trick	Example	Ask
Different denominators	“Route A has fewer complaints” (it has far fewer riders)	Per what?
Cherry-picked days	“Best week of the term”	Which days were left out?
Truncated axis	A bar chart starting at 80 makes a 5% gap look huge	Where does the axis start?

None of these needs a false number. Each one needs a question.

Four Questions for Any Interpretation

1. **What decision** is this number for?
2. **Which metric** fits that decision: average, peak, percentile, worst case?
3. **Which baseline, boundary, and time basis** is being used?
4. **Which assumption,** if wrong, would flip the conclusion?

Ask these before you accept a result, and before you present one.

Read It Twice

Route data (illustrative). On your own, on paper:

Average wait 6 min. 15% of riders wait over 12 min. 82% of trips on time. Peak: 300 riders per hour want a bus; capacity is 240.

1. Write one favourable sentence, as a dispatcher would say it.
2. Write one critical sentence, as a rider would say it.
3. Name one extra piece of data that would settle the argument.

Six minutes, on your own. Use only the numbers above.

Read It Twice: Debrief

Reader	A possible sentence	Data that would settle it
Dispatcher	“82% of trips are on time and the average wait is only 6 minutes.”	Waits by hour of day
Rider	“At the peak, 300 of us want a bus and only 240 can ride. One in seven waits over 12 minutes.”	Share of riders left behind at the peak

Both sentences are true. The extra data decides which problem matters more.

Reporting Honestly

RESULT

The answer

A plain sentence a reader can use.

RANGE

How far to trust it

Low to high, not one value.

TOP ASSUMPTION

What it rests on

The one that matters most.

WHAT WOULD CHANGE IT

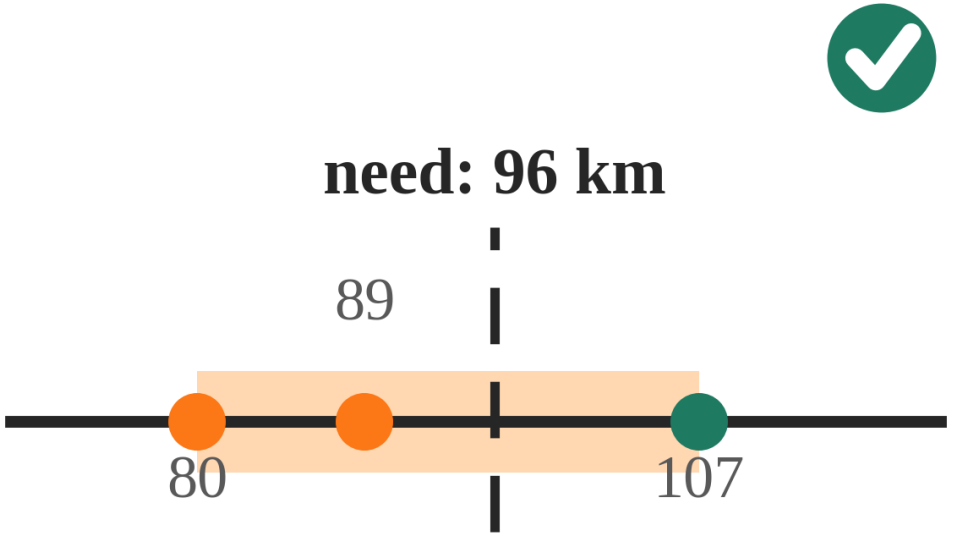
What to learn next

The data that would move it.

A good report gives the answer, how far to trust it, and what would change it.

Significant figures: Kosky et al., Ch. 2 (§2.5).

PART 3



result, range, assumption, what would change it

Putting It Together

One worked example, then your turn.

Worked Example: A Winter Day?

THE QUESTION

One bus, one charge

Can one bus cover a winter day of 96 km on one charge?

KNOWN

From earlier lectures

Usable energy 160 kWh. Daily need 96 km. Summer tests: 1.1, 1.3, and 1.6 kWh/km.

MISSING

No winter test

Cabin heating will raise energy use by an unknown amount.

We cannot measure it yet. We can still reason, and say how far to trust the result.

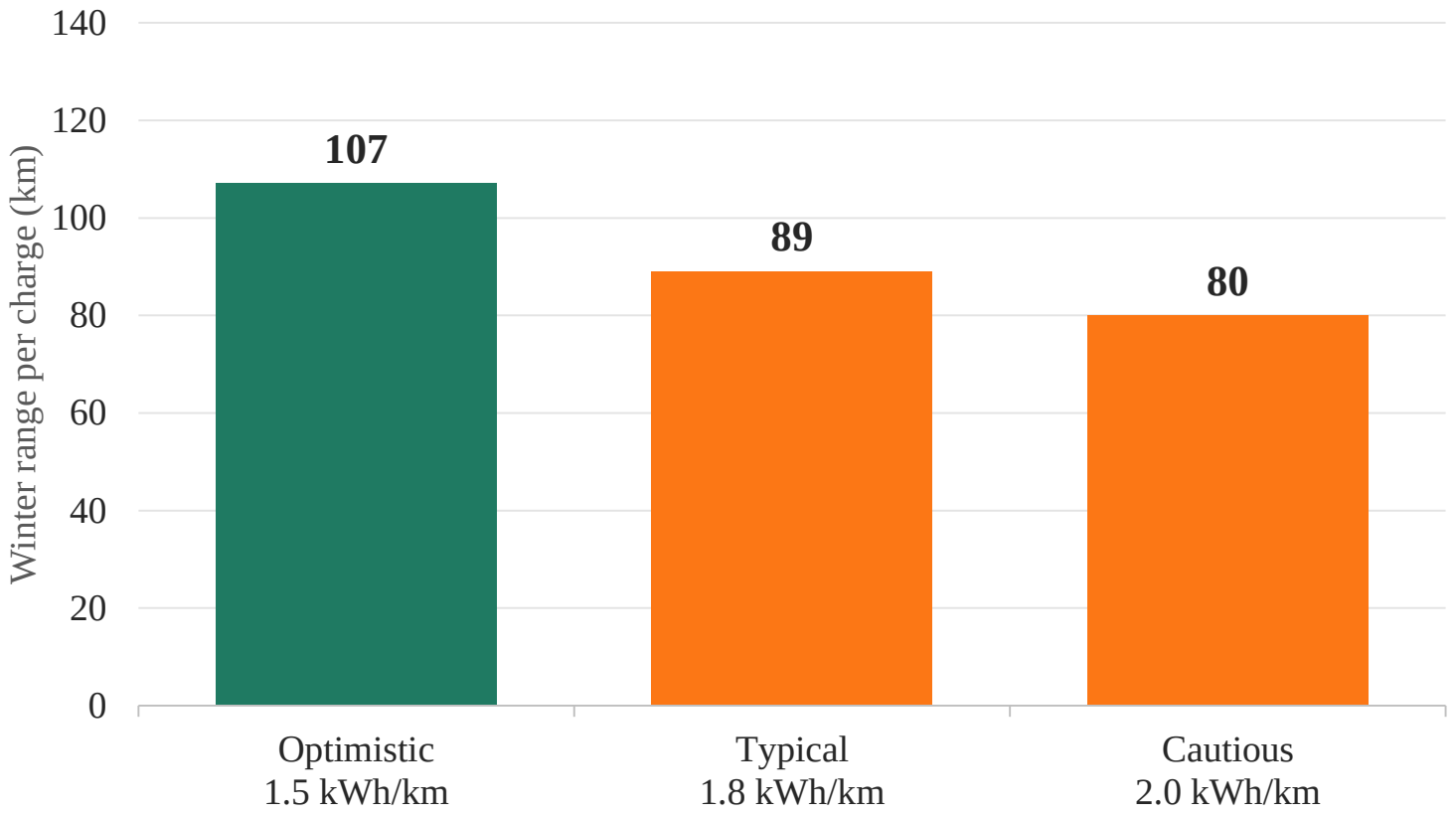
Fill the Gap with a Range

Quantity	Value or range	Source	Confidence
Usable energy	160 kWh	Datasheet value × usable share (Lecture 5)	Medium
Daily need	96 km	8 loops × 12 km	High
Summer energy use	1.1 to 1.6 kWh/km	Three test trips (n = 3)	Medium
Winter energy use	1.5 to 2.0, typical 1.8 kWh/km	Estimate by analogy: summer plus cabin heating	Low

Low confidence goes on the page. Hiding it would not make it go away.

Illustrative values.

Compute Low, Typical, and High



DAILY NEED

96 km

Passes: only the optimistic case.

Falls short: the typical and cautious cases.

Range = 160 kWh ÷ energy use per km.

Most of the range falls below the need. That is the finding.

Illustrative values.

Report It Honestly

RESULT

Probably not

A full winter day is unlikely: 89 km against 96 km.

RANGE

80 to 107 km

Only the optimistic case passes.

TOP ASSUMPTION

1.8 kWh/km

Winter energy use, with low confidence.

WHAT WOULD CHANGE IT

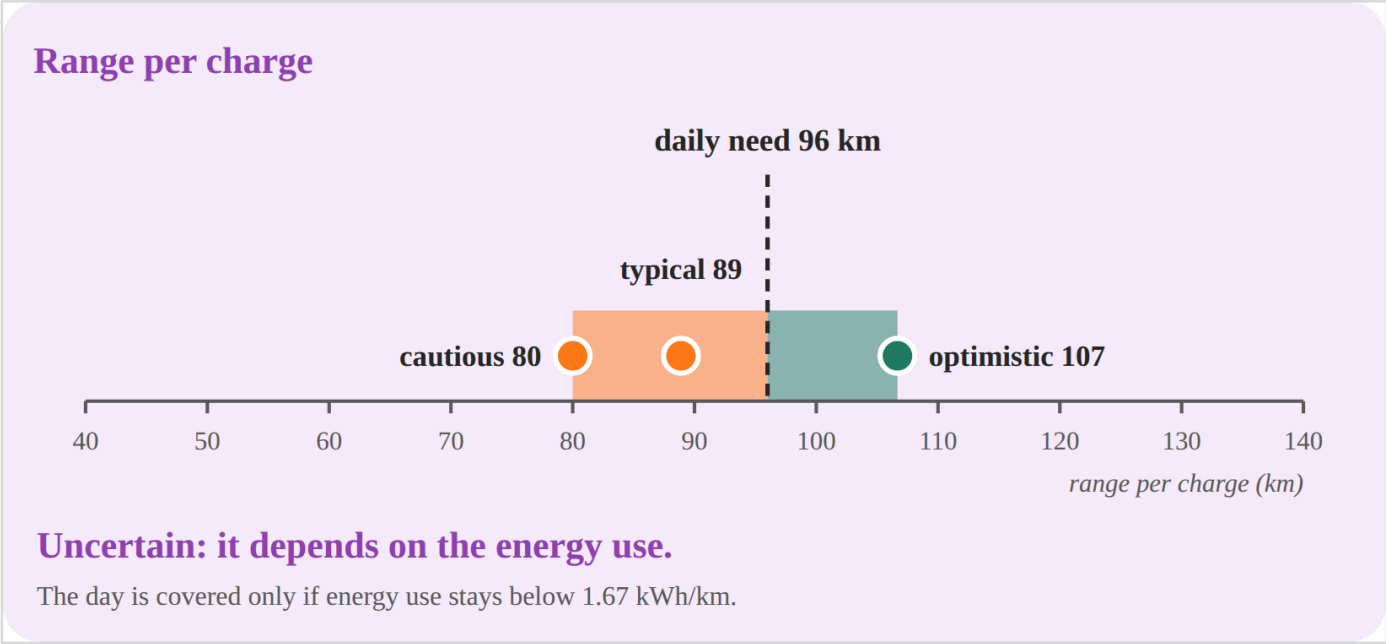
A winter test

One real winter trip, or a mid-day top-up.

Decision: plan a mid-day top-up, and collect winter data before committing.

Illustrative values.

Try It: The Winter Range Explorer



Scan, or open the link in the notes.

On your own, in about 3 minutes:

1. Choose “Summer tests”. Which trips clear the 96 km need?
2. Choose “Winter and aged battery”. How many km short is the best case?
3. From “Winter guess”, raise the usable share until the verdict changes. What share is needed?

Interactive page: https://claude.ai/artifact/NVrsyRbTJn6wF2q2fDDnTY Illustrative values.

Your Turn: Rooftop Solar for the Building

Can a 30 kW rooftop array cover the faculty building's daily electricity?

Given: the building averages 94 kW (Lecture 6). The array makes 3 to 5 kWh per kW per day.

1. Find the array's energy per day, as a range.
2. Find the building's energy per day.
3. Find the fraction covered, as a range.
4. Write one sentence: result, range, top assumption, what would change it.

Six minutes, on your own. Carry units in square brackets.

Your Turn: Answer Key

	Working	Answer
1	30 [kW] × 3 to 5 [kWh/kW-day]	90 to 150 kWh per day (typical 120)
2	94 [kW] × 24 [h]	about 2,260 kWh per day
3	90 ÷ 2,260 and 150 ÷ 2,260	about 4% to 7% (typical 5%)
4	Report	“The array covers about 4 to 7% of the building’s daily electricity, assuming 3 to 5 kWh per kW per day. A measured yield would change it.”

Both are true: a few percent of the building, but about one bus per day.

Illustrative values.

Key Takeaways

Name the problem

Too little, missing, noisy, unrepresentative, or wrong unit.

Know the source

Measure, look up, ask, estimate, or analogy. Write your trust.

Use ranges

Low, typical, high. Say n for small samples.

Triangulate

Estimate it more than one way and compare.

Read it twice

The metric and the baseline change the story.

Report honestly

Result, range, top assumption, what would change it.

Independent Study: Incomplete Data in Your System

1. Pick one quantity in your system where the data is incomplete, and name the kind of problem.
2. Write a low, typical, and high value, with a source and your confidence.
3. Show how the range changes one result.
4. Write one report sentence: result, range, top assumption, what would change it.

Nothing to hand in. This is working material for your own system, ready for the second half of the course.



Thank You — Questions?

COMING UP NEXT

Next week: Midterm Assessment

A paper-based assessment. Review the first seven weeks.